

From twilio.
rest import
TwilioRestClient.
This module will
import Twilio used
for sending text
messages.

import Py-
OWM module will
import PyOWM to
read humidity, and
import time module
for delays.

The above
two modules
have already been
installed and the
third module
comes pre-pack-
aged with Python.

You can check
all the different modules in Python
standard library or at [https://docs.
python.org/2/library/](https://docs.python.org/2/library/)

Functions used

Read current humidity level from
OpenWeatherMap website. The fol-
lowing function initialises with the
key obtained from OpenWeatherMap
website:

```
owm = pyowm.OWM('ENTER YOUR KEY')
```

The following function sets loca-
tion to Delhi, India:

```
observation = owm.weather_at_  
place('Delhi, IN')
```

The following function gets
weather status:

```
w = observation.get_weather()
```

The following function gets hu-
midity value:

```
humidity = w.get_humidity()
```

This section reads the current
humidity level in the specified area
and puts it in the variable humidity.

In the next section there is
print command for printing out
the read humidity level for testing
purpose. You can also read tem-
perature, wind condition and much
more with commands such as
w.get_temperature.

```

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... This program checks the humidity status on the internet for a specified
location and keeps alerting you the status through text messages...

from twilio.rest import TwilioRestClient
import pyowm
import time

while True:
    owm = pyowm.OWM('')
    observation = owm.weather_at_place('Delhi, IN')
    w = observation.get_weather()
    humidity = w.get_humidity()

    print humidity

    account_sid = ''
    auth_token = ''
    client = TwilioRestClient(account_sid, auth_token)

    if humidity>70:
        message = client.messages.create(body="Humidity exceeds set level",
            to="+91",
            from="+91")
        print message.sid

    else:
        message = client.messages.create(body="Humidity is below set level",
            to="+91",
            from="+91")
        print message.sid

    time.sleep(2)

```

Fig. 19: Test.py on Python GUI screen

Following code initialises Twilio
API for sending texts using account
SID and authorisation token:

```

account_sid = "ENTER YOUR SID"
auth_token = "ENTER YOUR TOKEN"
client = TwilioRestClient(account_sid,
auth_token)

```

Enter account SID and authorisa-
tion token you obtained from previ-
ous steps.

Following code sends 'Humidity
exceeds set level' text message' if
read humidity exceeds the set level
of 70 per cent:

```

if humidity>70:
    message = client.messages.create
    (body="Humidity exceeds set level",
    to="ENTER YOUR PHONE NUMBER",
    from="ENTER TWILIO NUMBER")
    print message.sid

```

The following section will
compare the read humidity value

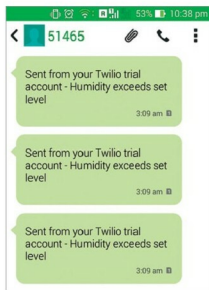


Fig. 20: Messages received on mobile phone

from OpenWeatherMap website to
the trigger of 70 per cent. You can
change the trigger as required.

If the read humidity exceeds 70
per cent, it will send a text mes-
sage to your mobile phone 'Humid-
ity exceeds set level'. If the read
humidity is below the set trigger, it
will send a text message 'Humidity
is below set level'.

Two-seconds time interval
setting is achieved by using time.
sleep(2) command.

Please note that the two-
seconds time is only for testing
purpose. If you want to read hu-
midity every two hours and receive
the text messages, replace 2 by
2 × 60 × 60 = 7200.

Running the program

Enter Python IDLE→File→Open.
You can also double-click on Text.
py file to run it directly on Win-
dows. The program can be termi-
nated with Ctrl + C in Interpreter
window.

Fig. 20 shows the alert messages
received on the mobile phone. **EFY**

EFY Note

The source code of
this project is
included in this
month's EFY DVD and
is also available for free
download at [source.
efymag.com](http://source.efymag.com)



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